

Formula Cheat Sheet

Chapter 3: Diode Models and Circuits

Diode Models

Three Diode Models

Model	ON Equivalent	OFF Equivalent	When to Use
Ideal	Short ($V_D = 0$)	Open ($I_D = 0$)	Quick intuition
Constant-Voltage	Battery $V_{D,on}$	Open ($I_D = 0$)	Most practical circuits
Exponential	$I_D = I_S e^{V_D/V_T}$	Same equation	Area/current details

Model Selection Rule

- Signal $\gg V_{D,on}$? $\rightarrow$ Ideal or Constant-Voltage
- Need exact currents or different diode areas? $\rightarrow$ Exponential
- Small perturbation around bias? $\rightarrow$ Small-Signal (r_d)

Core Equations

Exponential Diode Equation

$$I_D = I_S \exp\left(\frac{V_D}{V_T}\right) \quad (2.109)$$

$$V_T = \frac{kT}{q} \approx 26 \text{ mV at } 300 \text{ K}$$

Small-Signal Resistance

$$r_d = \frac{V_T}{I_D} = \frac{26 \text{ mV}}{I_D} \quad (3.58)$$

Small-Signal Current Change

$$\Delta I_D = \frac{I_D}{V_T} \Delta V_D \quad (3.49)$$

Small-Signal Voltage Change

$$\Delta V_D = \frac{V_T}{I_D} \Delta I_D = r_d \Delta I_D \quad (3.69)$$

Rectifier Formulas

Half-Wave Rectifier Average

$$V_{out,avg} = \frac{V_p}{\pi} \approx 0.318 V_p \quad (3.8)$$

Full-Wave Rectifier Average

$$V_{out,avg} = \frac{2V_p}{\pi} \approx 0.637 V_p \quad (1)$$

Half-Wave Ripple Voltage

$$V_R \approx \frac{V_p - V_{D,on}}{R_L C_1 f_{in}} \quad (3.80)$$

Full-Wave Ripple Voltage

$$V_R \approx \frac{V_p - 2V_{D,on}}{2 R_L C_1 f_{in}} \quad (3.94)$$

Ripple (Current Form)

$$V_R = \frac{I_L}{C_1 f_{in}} \quad (3.83)$$

Peak Diode Current

$$I_p \approx \frac{V_p}{R_L} \left(R_L C_1 \omega_{in} \sqrt{\frac{2V_R}{V_p} + 1} \right) \quad (3.92)$$

Max Reverse Voltage

Half-Wave: $V_{rev} \approx 2V_p$

Full-Wave: $V_{rev} \approx V_p$

Regulator Formulas**Zener Line Regulation**

$$\frac{v_{out}}{v_{in}} = \frac{r_D}{r_D + R_1} \quad (3.100)$$

Zener Load Regulation

$$\left| \frac{v_{out}}{i_L} \right| = (r_{D1} + r_{D2}) \parallel R_1 \quad (3.108)$$

Limiter & Divider Formulas**Voltage Divider (D ON, Ideal)**

$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in} \quad (3.9)$$

Voltage Divider (D ON, CV)

$$V_{out} = \frac{R_2 V_{in} + R_1 V_{D,on}}{R_1 + R_2} \quad (3.11)$$

Limiter Threshold

$$V_L = V_{D,on} + V_B$$

Capacitive Divider

$$\Delta V_{out} = \frac{C_1}{C_1 + C_2} \Delta V_{in} \quad (3.111)$$

Note: C_1 in numerator (opposite of resistive divider)

Voltage Doubler Output

$$V_{out} = 2V_p - 2V_{D,on}$$

Assume-and-Verify Quick Checklist

1. Assume ON or OFF for each diode
2. ON $\rightarrow$ short (ideal) or $V_{D,on}$ battery (CV). OFF $\rightarrow$ open circuit.
3. Solve with KVL/KCL
4. Verify: ON diode has $I_D > 0$? OFF diode has $V_D < V_{D,on}$?
5. Contradiction $\rightarrow$ flip assumption, re-solve